

Multiple Linear Regression

Experience $\longrightarrow$ Salary

Features $\rightarrow$ Salary

μ_2 Model.

Experience
Education
Skills
Age

Model

Salary

$$\text{Salary} = \text{Base salary} + \text{Experience effect} + \text{Education effect} + \text{Skill effect}$$

$$\text{Salary} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{Experience} + \beta_3 \text{Gender} + \dots$$

Target variable

intercept (base value)

feature 1
coefficient of x_1

feature 2
coefficient of x_2

feature 3
coefficient of x_3

$$\text{Salary} = 20,000 + 5000(\text{experience}) + 2000(\text{education})$$

$$Q = 20,000 + 5000(2) + 2000(\text{educatio})$$

$$Q = 30000 + 2500(\text{edw})$$

$$S' = 20000 + 5000(3) + 2000(\text{edukasi})$$

$$S = 35000 + 2000(\text{ednet})$$



1. advertisement $\rightarrow$ sales?
2. channel gives best ROI
3. new campaign budget, how many product we can expect to sell.

SALES

multiple predictors (TV, radio, newspaper)

features

TV	Appl. price	range	
radio	0-300	}	
newspaper	0-50		
computer	0-120		
profit	0-1000000000		

TV radio newspaper computer

230 38 75 1.12

0.9 .6

(matrix algebra)

OLS

$$R^2 \rightarrow \text{interpretation}$$

perfect prediction

1 → Excellent fit, the model explains about 90% of the variation in target variable
0.9 →

0.75 $\rightarrow$ Good fit, for many real-world problems.

0.5 $\longrightarrow$ Moderate fit

0.2 $\longrightarrow$ weak model simply predicting the average value over time.

negative $\rightarrow$ model simply predicting the
negative. $\rightarrow$ worse than the baseline itself

Fig. 1. —